Longest Subarray with Sum K

Solved

Difficulty: **Medium** Accuracy: **24.64%** Submissions: **880K+** Points: **4**

Given an array **arr[]** containing integers and an integer **k**, your task is to find the length of the longe subarray where the sum of its elements is equal to the given value k. If there is no subarray with su equal to k, return 0.

Examples:

Input: arr[] = [10, 5, 2, 7, 1, -10], k = 15
Output: 6
Explanation: Subarrays with sum = 15 are [5, 2, 7, 1], [10, 5] and [10, 5, 2, 7, 1, -10]. The length of the longest subarray with a sum of 15 is 6.

Input: arr[] = [-5, 8, -14, 2, 4, 12], k = -5
Output: 5
Explanation: Subarrays with sum = -5 are [-5] and [-5, 8, -14, 2, 4]. The length of the longest subarray with a sum of -5 is 5.

Input: arr[] = [10, -10, 20, 30], k = 5
Output: 0
Explanation: No subarray with sum = 5 is present in arr[].

Constraints:

$1 \leq arr.size() \leq 10^5$
$-10^4 \leq arr[i] \leq 10^4$
$-10^9 \leq k \leq 10^9$

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```java


// User function Template for Java


class Solution {
    public int longestSubarray(int[] arr, int k) {
        int maxl = 0;
        int sum = 0;
        HashMap<Integer, Integer> prefixSumMap = new HashMap<>();

        for (int i = 0; i < arr.length; i++) {
            sum += arr[i];
            if (sum == k) {
                maxl = i + 1;
            }
            if (prefixSumMap.containsKey(sum - k)) {
                maxl = Math.max(maxl, i - prefixSumMap.get(sum - k));
            }
            if (!prefixSumMap.containsKey(sum)) {
                prefixSumMap.put(sum, i);
            }
        }
        return maxl;
    }
```

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